Map Projection Theory

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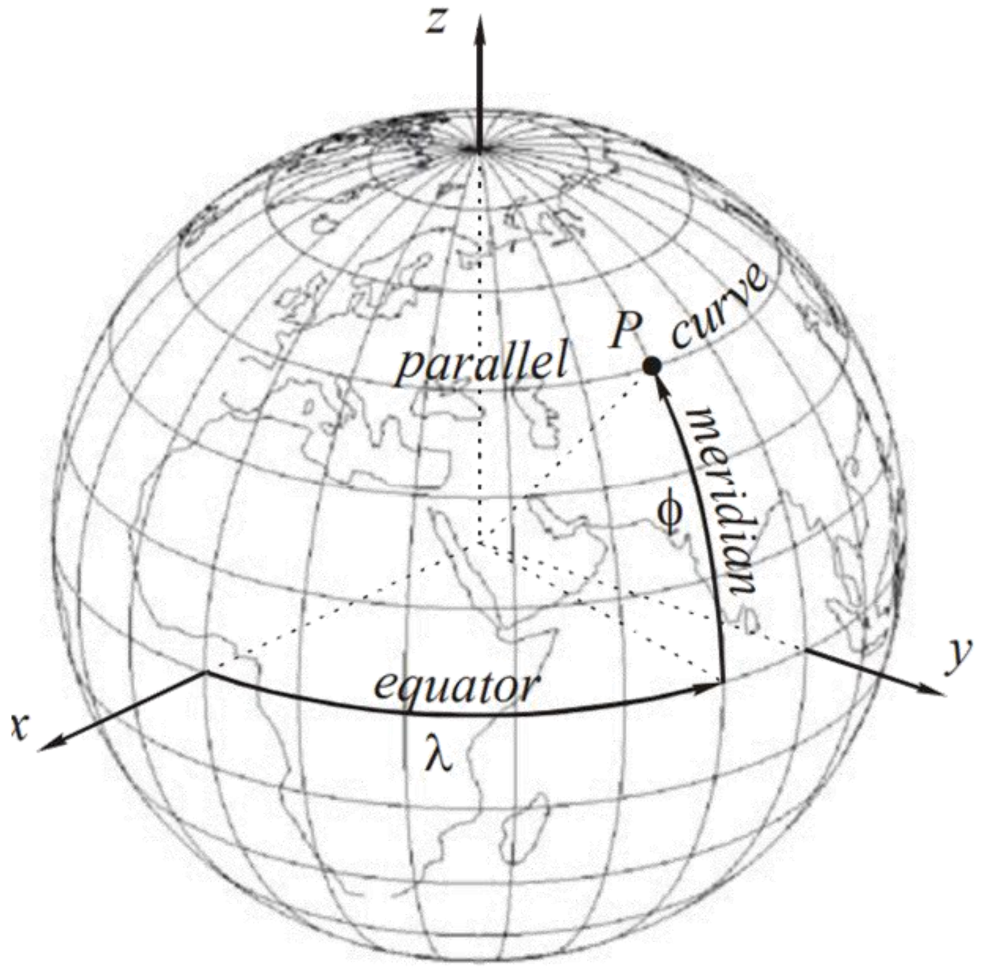
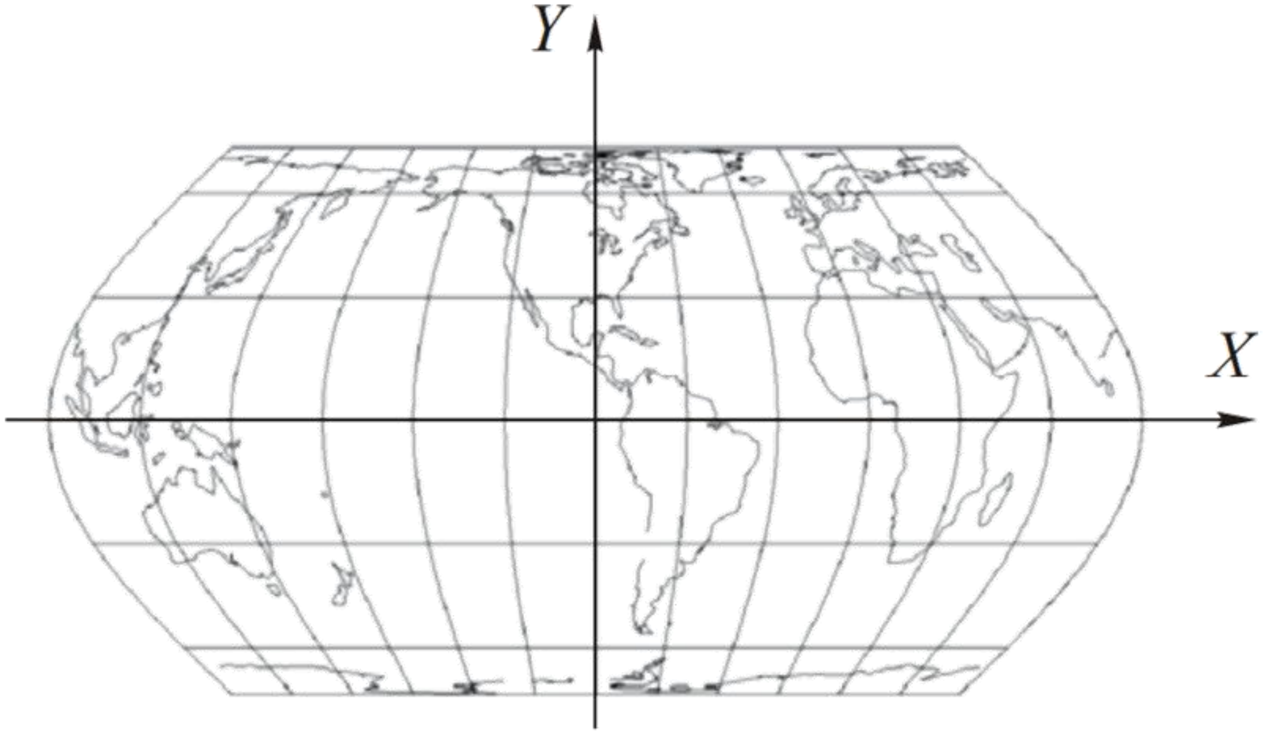
Geograhic Coord.



Projected Coord.



x,y,z Cartesian vs ϕ,λ curvilinear

$$x = f_1(\phi, \lambda) = R\cos\phi\cos\lambda$$

$$y = f_2(\phi, \lambda) = R\cos\phi\sin\lambda$$

$$z = f_3(\phi, \lambda) = R\sin\phi$$

map projection

$$X = F_1(U,V)$$
$$Y = F_2(U,V)$$
$$Z = F_3(U,V)$$

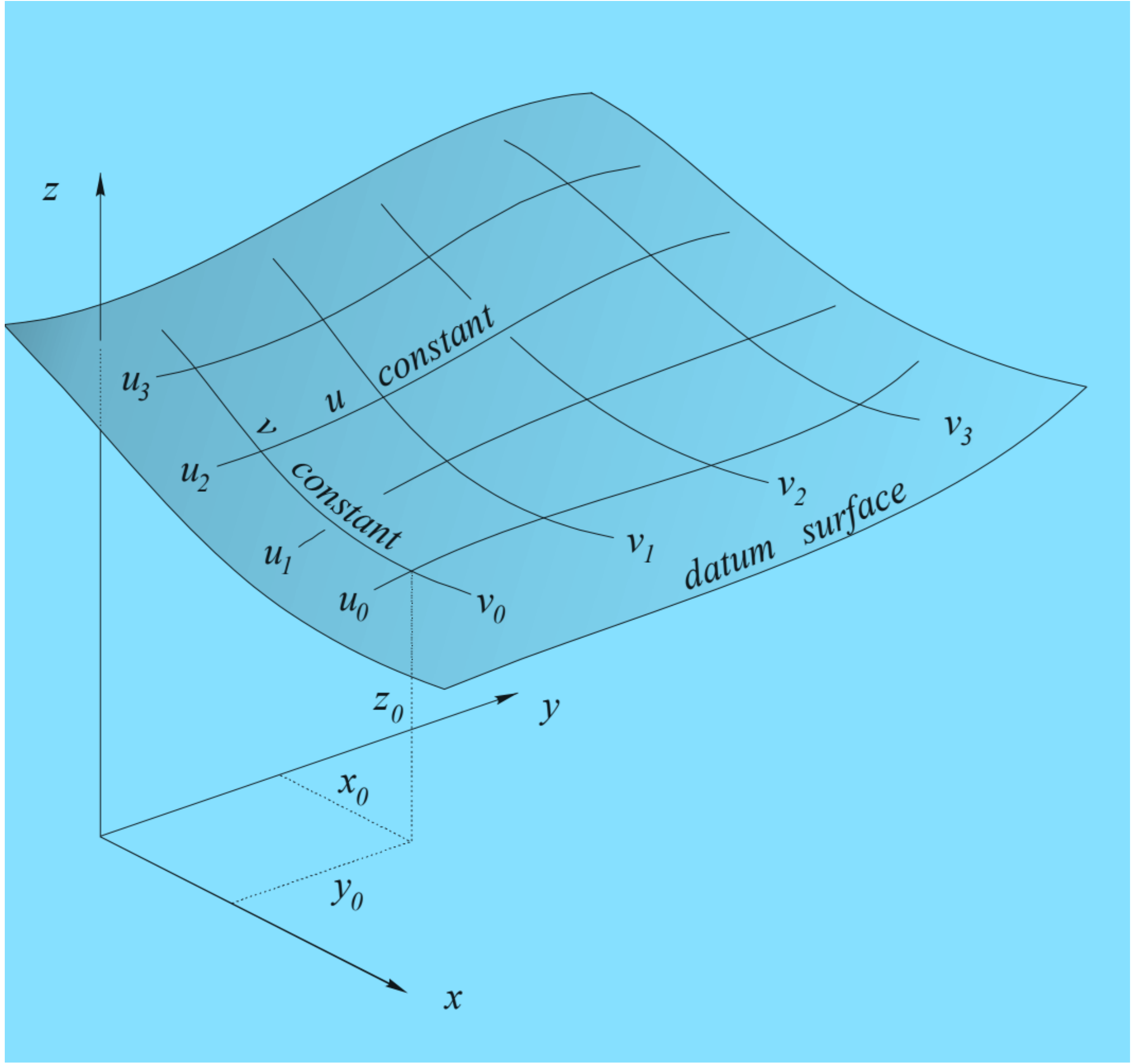
$$X = \overline{f}_1(\phi,\lambda)$$
$$Y = \overline{f}_2(\phi,\lambda)$$

Datum Surface

$$x = f_1(u, v)$$

$$y = f_2(u, v)$$

$$z = f_3(u, v)$$

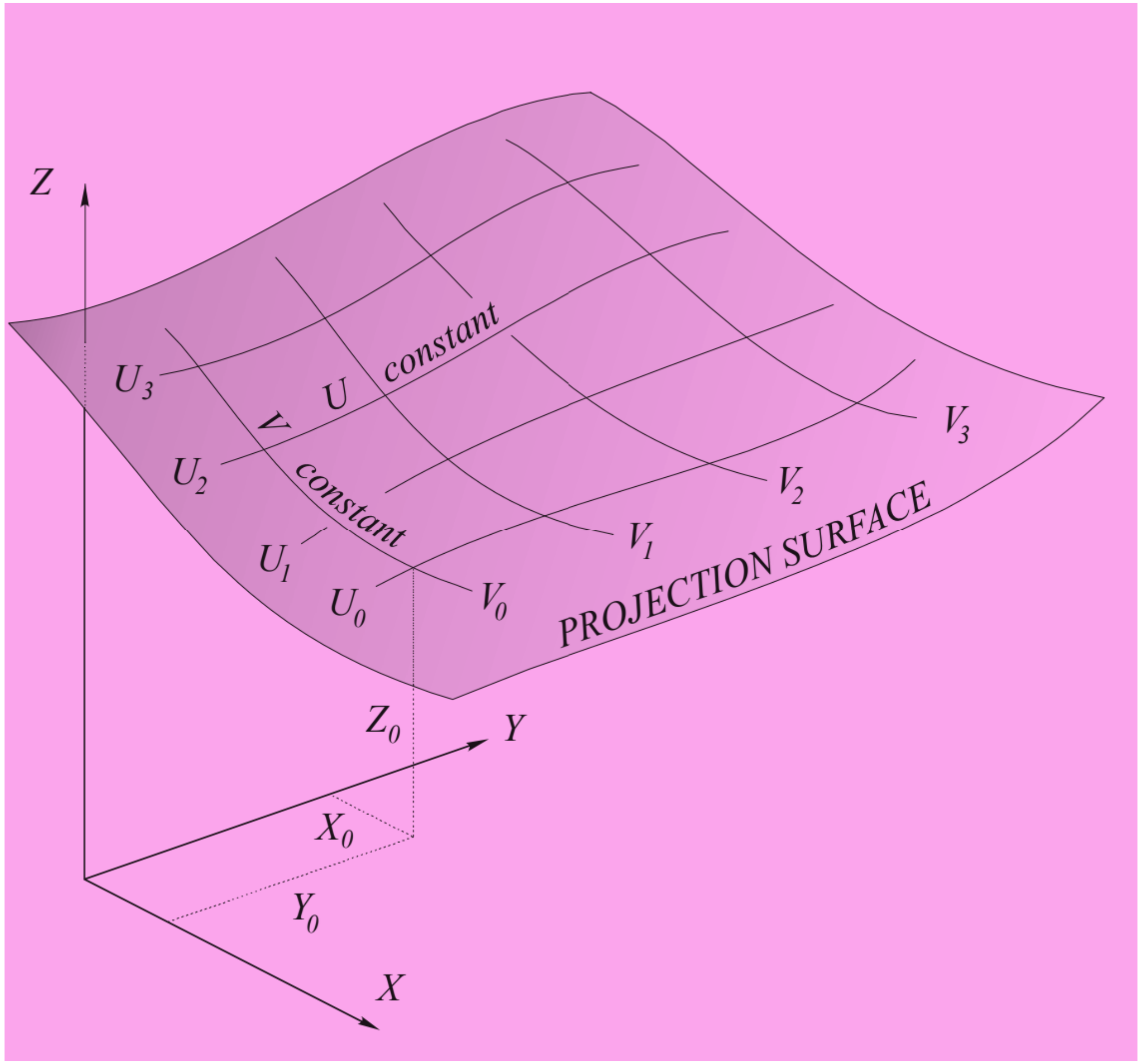


Projection Surface

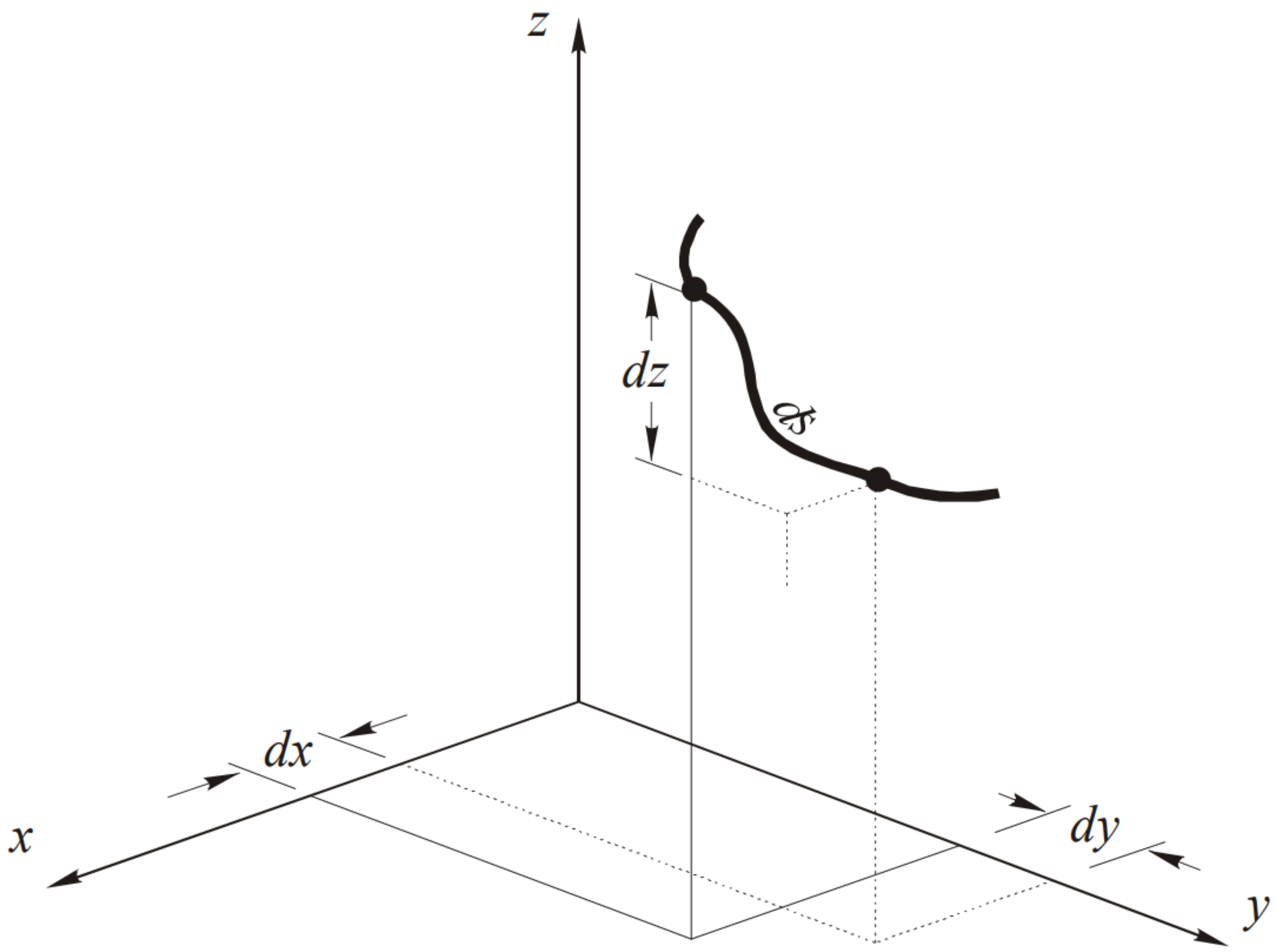
$$X = F_1(U,V)$$

$$Y = F_2(U,V)$$

$$Z = F_3(U,V)$$



THE GAUSSIAN FUNDAMENTAL QUANTITIES ON THE DATUM SURFACE



From differential geometry, the square of the length of a <u>differentially small</u> part of a curve on the datum surface is

$$ds^2 = dx^2 + dy^2 + dz^2 \tag{3.1}$$

$$dx = \frac{\partial x}{\partial u} du + \frac{\partial x}{\partial v} dv$$
$$dy = \frac{\partial y}{\partial u} du + \frac{\partial y}{\partial v} dv$$
$$dz = \frac{\partial z}{\partial u} du + \frac{\partial z}{\partial v} dv$$

$$\begin{aligned} ds^2 = & \left(\frac{\partial x}{\partial u}\right)^2 du^2 + \left(\frac{\partial x}{\partial v}\right)^2 dv^2 + 2\frac{\partial x}{\partial u}\frac{\partial x}{\partial v} du\,dv \\ & + \left(\frac{\partial y}{\partial u}\right)^2 du^2 + \left(\frac{\partial y}{\partial v}\right)^2 dv^2 + 2\frac{\partial y}{\partial u}\frac{\partial y}{\partial v} du\,dv \\ & + \left(\frac{\partial z}{\partial u}\right)^2 du^2 + \left(\frac{\partial z}{\partial v}\right)^2 dv^2 + 2\frac{\partial z}{\partial u}\frac{\partial z}{\partial v} du\,dv \end{aligned}$$

$$\begin{aligned} ds^2 = & \left\{\left(\frac{\partial x}{\partial u}\right)^2 + \left(\frac{\partial y}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial u}\right)^2\right\} du^2 \\ & + \left\{\frac{\partial x}{\partial u}\frac{\partial x}{\partial v} + \frac{\partial y}{\partial u}\frac{\partial y}{\partial v} + \frac{\partial z}{\partial u}\frac{\partial z}{\partial v}\right\} 2du\,dv \\ & + \left\{\left(\frac{\partial x}{\partial v}\right)^2 + \left(\frac{\partial y}{\partial v}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right\} dv^2 \end{aligned}$$

Gaussian Fundamental Quantities

$$ds^2 = e\,du^2 + 2f\,du\,dv + g\,dv^2$$

$$e = \left(\frac{\partial x}{\partial u}\right)^2 + \left(\frac{\partial y}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial u}\right)^2$$

$$f = \frac{\partial x}{\partial u}\frac{\partial x}{\partial v} + \frac{\partial y}{\partial u}\frac{\partial y}{\partial v} + \frac{\partial z}{\partial u}\frac{\partial z}{\partial v}$$

$$g = \left(\frac{\partial x}{\partial v}\right)^2 + \left(\frac{\partial y}{\partial v}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2$$

Elemental distance on the surface of the spherical Earth

$$x = f_1(\phi, \lambda) = R\cos\phi\cos\lambda$$
$$y = f_2(\phi, \lambda) = R\cos\phi\sin\lambda$$
$$z = f_3(\phi, \lambda) = R\sin\phi$$

$$e = \left(\frac{\partial x}{\partial \phi}\right)^2 + \left(\frac{\partial y}{\partial \phi}\right)^2 + \left(\frac{\partial z}{\partial \phi}\right)^2$$
$$f = \frac{\partial x}{\partial \phi}\frac{\partial x}{\partial \lambda} + \frac{\partial y}{\partial \phi}\frac{\partial y}{\partial \lambda} + \frac{\partial z}{\partial \phi}\frac{\partial z}{\partial \lambda}$$
$$g = \left(\frac{\partial x}{\partial \lambda}\right)^2 + \left(\frac{\partial y}{\partial \lambda}\right)^2 + \left(\frac{\partial z}{\partial \lambda}\right)^2$$

Gaussian Fundamental Quantities for Sphere

$$\frac{\partial x}{\partial \phi} = -R\sin\phi\cos\lambda \qquad \frac{\partial x}{\partial \lambda} = -R\cos\phi\sin\lambda$$

$$\frac{\partial y}{\partial \phi} = -R\sin\phi\sin\lambda \qquad \frac{\partial x}{\partial \lambda} = R\cos\phi\cos\lambda$$

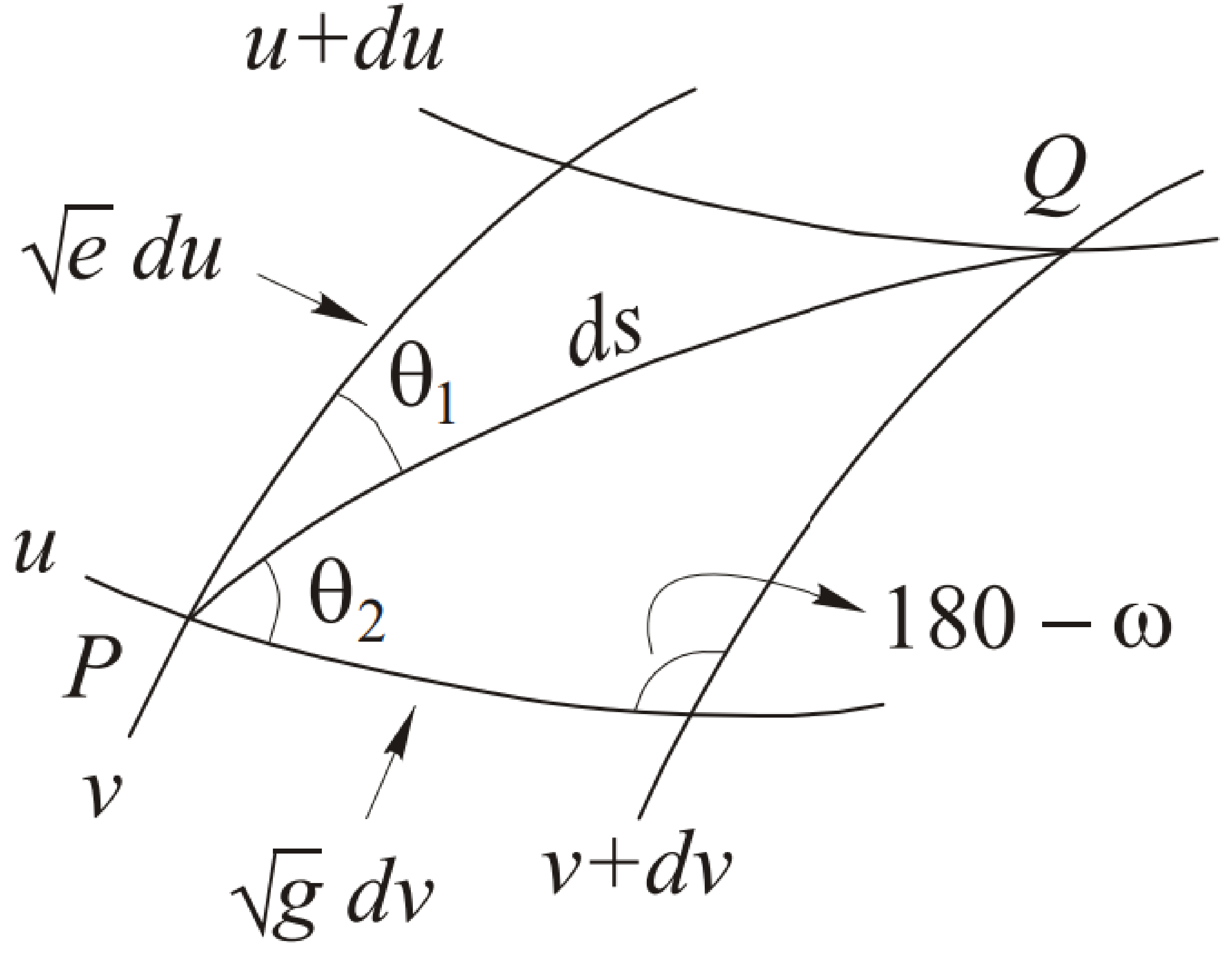
$$\frac{\partial x}{\partial \phi} = R\cos\phi \qquad \frac{\partial x}{\partial \lambda} = 0$$

$$e = R^2$$

$$f = 0$$

$$g = R^2\cos^2\phi$$

PARALLELOGRAM ON THE DATUM SURFACE



$$\omega = \theta_1 + \theta_2$$

distance along parametric curves

$$
\begin{aligned}
ds_v^2 &= e\,du^2 + 2f\,du\,dv + g\,dv^2 \\
&= e\,du^2 + 2f\,du(0) + g(0)^2 \\
&= e\,du^2
\end{aligned}
$$

$$ds_v = \sqrt{e}\,du$$

$$
\begin{aligned}
ds_u^2 &= e\,du^2 + 2f\,du\,dv + g\,dv^2 \\
&= e(0)^2 + 2f(0)dv + g\,dv^2 \\
&= g\,dv^2
\end{aligned}
$$

$$ds_u = \sqrt{g}\,dv$$

angle between parametric curves on the datum surface

$$ds^2 = e\,du^2 + g\,dv^2 - 2\left(\sqrt{e}\,du\right)\left(\sqrt{g}\,dv\right)\cos\left(180 - \omega\right)$$

$$= e\,du^2 + g\,dv^2 + 2\sqrt{eg}\,du\,dv\cos\omega$$

$$\cos\omega = \frac{f}{\sqrt{eg}} \qquad \sin\omega = \frac{j}{\sqrt{eg}}$$

area on the datum surface

$$
\begin{aligned}
da &= \left(\sqrt{e}\,du\right)\left(\sqrt{g}\,dv\right)\sin\omega \\
&= \sqrt{eg}\,\sin\omega\,du\,dv
\end{aligned}
$$

$$da = j\,du\,dv$$

Note: (i) The Gaussian Fundamental Quantities pertaining to the datum surface (x,y,z as functions of u and v) are denoted by e, f and g.

(ii) The Gaussian Fundamental Quantities pertaining to the projection surface (X,Y,Z as functions of U and V) are denoted by $\bar{E}$, $\bar{F}$ and $\bar{G}$.

(iii) The Gaussian Fundamental Quantities pertaining to the projection surface (X,Y,Z as functions of u and v) are denoted by E, F and G.

Simple comparison

Symbols	Surface	Parameters used
e, f, g	Datum surface, such as an ellipsoid	u, v, commonly ϕ, λ
$\bar{E}, \bar{F}, \bar{G}$	Projection surface	Its own parameters U, V
E, F, G	Projection surface	Datum parameters u, v

Main idea

$$\boxed{\bar{E}, \bar{F}, \bar{G}}$$

and

$$\boxed{E, F, G}$$

refer to the **same projection surface**, but they use different coordinate systems.

A simple analogy is:

- $\bar{E}, \bar{F}, \bar{G}$: describe the map using map coordinates U, V.
- E, F, G: describe the same map using geographic coordinates ϕ, λ.
- e, f, g: describe the original Earth surface using ϕ, λ.

This allows us to compare

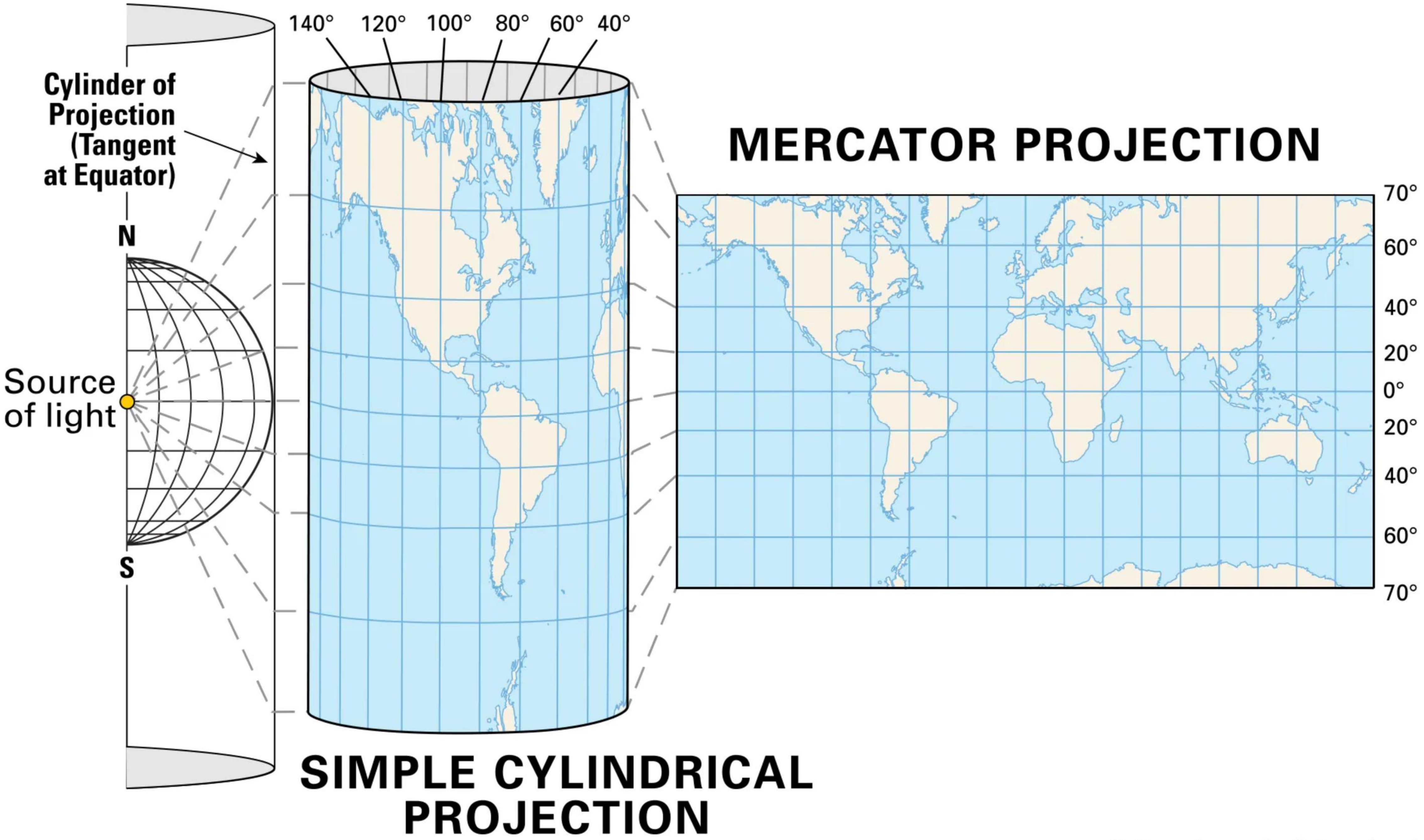
$$ds^2 = e\,du^2 + 2f\,du\,dv + g\,dv^2$$

on the Earth with

$$dS^2 = E\,du^2 + 2F\,du\,dv + G\,dv^2$$

on the map, which is the basis for studying **map scale, angular distortion, and map projection deformation**.

Gaussian Fundamental Quantities and Mercator


140°
120°
100°
80°
60°
40°
Cylinder of Projection (Tangent at Equator)
N
S
Source of light
MERCATOR PROJECTION
70°
60°
40°
20°
0°
20°
40°
60°
70°
SIMPLE CYLINDRICAL PROJECTION

เปรียบเทียบ e, f, g กับ E, F, G

ตัวอย่างที่ง่ายที่สุด: Simple Mercator บนทรงกลม

โลกทรงกลม
พารามิเตอร์ ϕ, λ

→

ฉายเป็นแผนที่
$X(\phi,\lambda)$, $Y(\phi,\lambda)$

→

เปรียบเทียบระยะ
หา scale factor

แนวคิดหลัก: e,f,g = บนโลก ส่วน E,F,G = บนแผนที่

นิยามสั้น ๆ

ใช้พารามิเตอร์เดียวกัน $u = \phi$, $v = \lambda$ เพื่อเปรียบเทียบโลกกับแผนที่

บนผิวโลก: e, f, g

$$ds^2 = e\, d\phi^2 + 2f\, d\phi\, d\lambda + g\, d\lambda^2$$

บอกระยะเล็ก ๆ บนทรงกลม/ทรงรีโลก

เหมาะสำหรับอธิบาย “ของจริงบนโลก”

บนแผนที่: E, F, G

$$dS^2 = E\, d\phi^2 + 2F\, d\phi\, d\lambda + G\, d\lambda^2$$

บอกระยะเล็ก ๆ หลังฉายเป็นแผนที่

ใช้เปรียบเทียบเพื่อหาความผิดเพี้ยน

วิธีจำง่าย : ตัวเล็ก = โลก, ตัวใหญ่ = แผนที่

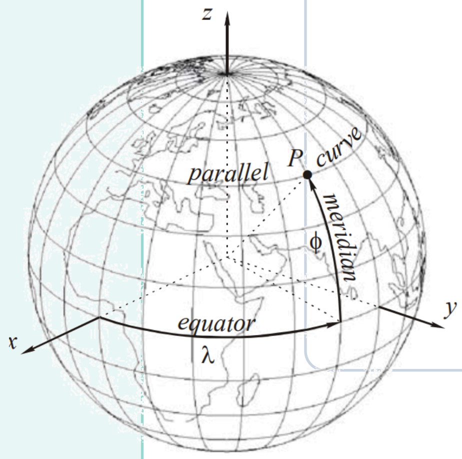
1) ผิวโลกทรงกลม: e, f, g

สำหรับ Simple Mercator ใช้ตัวอย่างโลกเป็นทรงกลมรัศมี R

พิกัดบนทรงกลม

$$x = R\cos\phi\cos\lambda$$
$$y = R\cos\phi\sin\lambda$$
$$z = R\sin\phi$$

จากอนุพันธ์ตาม ϕ และ λ



ผลลัพธ์บนผิวโลก

$$e = R^2$$
$$f = 0$$
$$g = R^2\cos^2\phi$$

$$ds^2 = R^2\,d\phi^2 + R^2\cos^2\phi\,d\lambda^2$$

2) แผนที่ Simple Mercator: E, F, G

สมการฉายแผนที่แบบทรงกลม

สมการฉาย

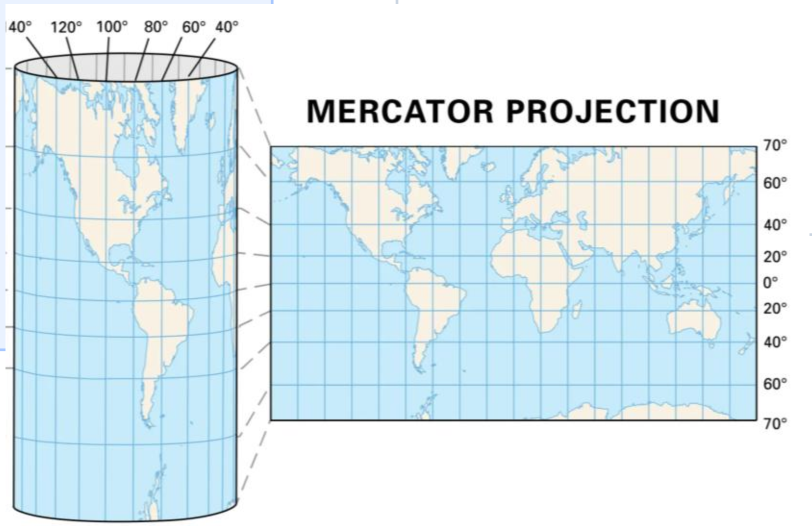
$$X = R\lambda$$
$$Y = R \ln \tan(\pi/4 + \phi/2)$$
$$Z = 0$$

แผนที่เป็นระนาบ จึงใช้ X,Y เท่านั้น

อนุพันธ์ที่ใช้

$$X_\phi = 0, \quad Y_\phi = R \sec\phi$$
$$X_\lambda = R, \quad Y_\lambda = 0$$

$$E = R^2 \sec^2\phi$$
$$F = 0$$
$$G = R^2$$



3) เปรียบเทียบโลกกับแผนที่

นำ e,f,g และ E,F,G มาเทียบกันเพื่อหามาตราส่วน

ปริมาณ	บนโลกทรงกลม	บนแผนที่ Mercator
e หรือ E	$e = R^2$	$E = R^2 \sec^2\phi$
f หรือ F	$f = 0$	$F = 0$
g หรือ G	$g = R^2 \cos^2\phi$	$G = R^2$

$$h = \sqrt{(E/e)} = \sec\phi$$

$$k = \sqrt{(G/g)} = \sec\phi$$

มาตราส่วนสองทิศเท่ากัน → Mercator รักษามุม (conformal)

Simple Mercator เป็นตัวอย่างเริ่มต้นที่ง่ายที่สุด

1. โลก

$$e = R^2$$
$$f = 0$$
$$g = R^2 \cos^2\phi$$

คำนวณง่ายจากทรงกลม

2. แผนที่

$$E = R^2 \sec^2\phi$$
$$F = 0$$
$$G = R^2$$

สมการ Mercator สั้น

3. ความหมาย

$$h = k = \sec\phi$$

รักษามุม
แต่ขยายมากเมื่อ ϕ สูง

ประโยคจำ: “ตัวเล็กวัดบนโลก — ตัวใหญ่วัดบนแผนที่”

Ex04 : Gaussian Fundamental Quantities for Spherical Mercator Projection

Derivation of Gaussian Fundamental Quantities

Dr.-Ing. Phisan Santitamnont and AI-LLM

Coordinate System Definitions

Let R be the radius of the spherical Earth.

We define the datum surface parameters u and v as:

- u = geodetic latitude (ϕ), where $u \in [-\pi/2, \pi/2]$
- v = longitude (λ), where $v \in [-\pi, \pi]$

The 3D Cartesian coordinates (x, y, z)on the datum surface (sphere) as functions of u, v are:

$$x = R\cos u\cos v$$
$$y = R\cos u\sin v$$
$$z = R\sin u$$

(i) Datum Surface Quantities: *e, f, g*

$$r(u,v) = (x,y,z)$$

Let the position vector be . We first calculate the partial derivatives and : r_u r_v

$$r_u = -R\sin u\cos v, \quad -R\sin u\sin v, \quad R\cos u$$
$$r_v = -R\cos u\sin v, \quad R\cos u\cos v, \quad 0$$

Now, calculate the First Fundamental Form coefficients via dot products:

$$e = r_u \cdot r_u = (-R\sin u\cos v)^2 + (-R\sin u\sin v)^2 + (R\cos u)^2$$
$$e = R^2\sin^2 u(\cos^2 v + \sin^2 v) + R^2\cos^2 u$$
$$e = R^2(\sin^2 u + \cos^2 u) = R^2$$

(i) Datum Surface Quantities: *e, f, g* (cont.)

Continuing the calculations for f and g:

$$f = r_u \cdot r_v$$
$$f = (R^2 \sin u \cos u \sin v \cos v) - (R^2 \sin u \cos u \sin v \cos v) + 0$$
$$f = 0$$

$$g = r_v \cdot r_v$$
$$g = (-R \cos u \sin v)^2 + (R \cos u \cos v)^2 + 0$$
$$g = R^2 \cos^2 u (\sin^2 v + \cos^2 v)$$
$$g = R^2 \cos^2 u$$

Result: $e = R^2, \quad f = 0, \quad g = R^2 \cos^2 u$

(ii) Cartesian Plane Quantities: $\bar{E}, \bar{F}, \bar{G}$

The standard projection surface for the developed Mercator map is a 2D Cartesian plane (unrolled cylinder). Let U be Easting and V be Northing.

$$X = U$$
$$Y = V$$
$$Z = 0$$

Let position vector $\mathbf{R}(U, V) = (U, V, 0)$. The partial derivatives are:

$$R_U = (1, 0, 0), \quad R_V = (0, 1, 0)$$

Calculating the fundamental quantities:

$$\bar{E} = R_U \cdot R_U = 1^2 + 0 + 0 = 1$$
$$\bar{F} = R_U \cdot R_V = 0$$
$$\bar{G} = R_V \cdot R_V = 0 + 1^2 + 0 = 1$$

(iii) Projection Surface Quantities: *E, F, G*

We evaluate the projection plane coordinates (X, Y, Z) using the forward spherical Mercator mapping equations in terms of the datum coordinates u, v:

$$X(u, v) = Rv$$
$$Y(u, v) = R \ln \tan\left(\frac{\pi}{4} + \frac{u}{2}\right) = R \ln(\sec u + \tan u)$$
$$Z(u, v) = 0$$

Let $\mathbf{R}(u, v) = (X(u, v), Y(u, v), 0)$. Calculate partial derivatives for u and v:

For u(latitude):

$$X_u = 0$$
$$Y_u = \frac{d}{du}[R \ln(\sec u + \tan u)] = R\left(\frac{\sec u \tan u + \sec^2 u}{\sec u + \tan u}\right) = R \sec u$$
$$\mathbf{R}_u = (0, R \sec u, 0)$$

(iii) Projection Surface Quantities: *E, F, G* (cont.)

Derivatives for v (longitude):

$$X_v = R, \quad Y_v = 0$$
$$R_v = (R, \quad 0, \quad 0)$$

Now calculate E, F, G:

$$E = R_u \cdot R_u = 0^2 + (R\sec u)^2 + 0^2 = R^2 \sec^2 u$$
$$F = R_u \cdot R_v = (0)(R) + (R\sec u)(0) + 0 = 0$$
$$G = R_v \cdot R_v = R^2 + 0^2 + 0^2 = R^2$$

Result: $E = R^2 \sec^2 u, \quad F = 0, \quad G = R^2$

Gaussian Fundamental for Spherical Mercator

Spherical Datum x,y,z *vs* **atitude** u **and longitude** v	**e**	**f**	**g**
$projection\ surface$ (Easting, Northing)	$\bar{E}$	$\bar{F}$	$\bar{G}$
Projection surface latitude (ϕ) and longitude (λ),	E	F	G

Ex.	รหัสนิสิต ชื่อ-นามสกุล	99	

Verification of Conformality

Because the Mercator projection is strictly **conformal**, the scale factor m must be equal in all directions.

We verify this mathematically by checking that the fundamental quantities of the projection surface are strictly proportional to those of the datum surface by a factor of m^2:

$$\frac{E}{e} = \frac{R^2 \sec^2 u}{R^2} = \sec^2 u$$

$$\frac{G}{g} = \frac{R^2}{R^2 \cos^2 u} = \sec^2 u$$

Since $E/e = G/g = \sec^2 u$ and $F = f = 0$, the derivation confirms the known point scale factor for the spherical Mercator projection:

$$m = \sec u$$